

Article

A Two-Tier Planning Approach for Hybrid Energy Storage Systems Considering Grid Power Flexibility in New Energy High-Penetration Grids

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Abstract

This paper proposes a flow battery-lithium-ion battery hybrid energy storage system (HESS) bi-level optimization planning method to address flexibility supply-demand balance challenges in regional power grids with high renewable penetration at 220 kV and above voltage levels. The method establishes a planning-operation coordination framework: Upper-level planning minimizes total lifecycle investment and operation-maintenance costs; Lower-level operation incorporates multiple constraints including flexibility gap penalties, voltage fluctuations, and line losses, overcoming single-timescale limitations. The approach enhances global search capability through the Improved Weighted Average Algorithm (IWAA) and optimizes power allocation accuracy using adaptive Variational Mode Decomposition (VMD). Validation using grid data from Southwest China demonstrates significant improvements across five comparative schemes. Results show substantial reductions in total investment costs, penalty costs, voltage fluctuations, and line losses compared to benchmark solutions, enhancing grid power supply stability and verifying the effectiveness of the model and algorithm.

Keywords: Bi-level optimization; HESS; renewable energy integration; VMD; weighted adaptive algorithm

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1. Introduction

In recent years, with the increasing integration of renewable energy into power grids, the spatio-temporal mismatch between renewable energy generation and load demand has become increasingly prominent. Issues such as wind and solar curtailment as well as load shedding pose significant challenges to the secure operation of regional grids [1]. Flexibility supply-demand balance refers to the extent to which the flexibility provided by various resources exceeds the flexibility requirement, at any moment, over any timescale, and in any direction, exceeding a permissible threshold. As the penetration of renewable energy continues to rise, flexibility is becoming the core and critical aspect of

system operational characteristics [2]. Confronted with the challenge of renewable energy fluctuations spanning second-to-day timescales, short-term energy storage can safeguard instantaneous system security and stability, while long-term energy storage significantly reduces wind/solar curtailment and load shedding. Consequently, investigating hybrid short- and long-term energy storage systems (ESSs) to overcome the temporal limitations of single-type storage and achieve flexibility supply-demand matching across critical timescales is crucial to ensuring both the secure operation and enhanced flexibility of high-penetration renewable energy systems.

As a key approach to enhancing renewable energy accommodation capacity in high-penetration systems, wind-PV-storage integrated system planning has been extensively studied regarding the collaborative configuration of flexibility resources. Zipeng Liang et al. proposed a planning framework based on a three-level structure involving bidirectional converters to enhance the reliability and robustness of hybrid AC/DC microgrid clusters with high renewable energy penetration [3]. Shutan Wu et al. considered the spatiotemporal flexible resources and multi-network coupling effects, and proposed a cyber-physical integrated planning model for distribution networks. They established a three-level planning model to iteratively solve the planning problem. Case studies verified that the proposed scheme can effectively alleviate the issue of voltage surplus [4]. Dundun Li et al. considered that increased wind integration and coal-plant retirements would heighten operational pressures on power systems, proposing a three-level robust ESS planning model to accommodate uncertainties in wind power investment and coal unit retirements [5]. Zongxiang Lu et al. established a quantitative index system for power system flexibility and developed a collaborative flexibility planning framework integrating source-grid-load-storage resources [6]. Xiaorong Zhu et al. explored a coordinated planning method for energy storage and Soft Open Points (SOPs) in distribution networks, optimizing resource allocation using multi-timescale flexibility indices [7]. Lijun Liu et al. introduced a joint planning model for long- and short-duration storage, employing a deep clustering algorithm to characterize annual chronological features [2]. Deukyoung Lee et al. proposed a novel integrated planning method for generators, transmission lines, and ESSs, using a two-stage Benders decomposition approach [8]. Yanda Huo et al. developed a lightweight data-driven planning model and quantified the demand function for HESS in new power systems based on production simulation [9]. However, the flexibility resource synergy mechanism in the above literature is imperfect and does not fully integrate the dynamic response characteristics of multi-dimensional resources of source, network, load and storage.

Energy storage systems possess the capability for bidirectional energy flow, offering advantages such as energy shifting capability, rapid response, and flexible deployment. These attributes enable ESS to effectively alleviate power supply pressure during peak load periods, optimize power flow distribution, and enhance grid operational efficiency. In recent years, extensive research on hybrid energy storage has been conducted. Rukang Wang et al. addressed the challenge of wind-PV farm output fluctuations impacting grid stability by proposing a hybrid ESS strategy for mitigating composite power fluctuations, specifically considering the complementary characteristics between wind and PV power generation [10]. Xin Wang et al. leveraged Multi-Agent Deep Reinforcement Learning (MADRL); they first analyzed the relationship between output power and wake effects, subsequently establishing a power smoothing control model for multi-wind turbines and energy storage systems [11]. Yutong Li proposed a method termed Principal Component Analysis-based Incomplete Data Equivalence to build data-driven energy storage dispatch models under conditions of incomplete data [12]. Fatma Gülşen Erdiñç introduced an innovative approach for real-time energy management in residential communities, simultaneously considering the utilization of solar PV and ESS while

targeting cost minimization as the optimization objective [13]. Notably, the existing literature primarily focuses on wind-storage or PV-storage integrated systems, exploring the construction of energy storage primarily to mitigate power fluctuations from wind or solar sources. Relatively limited attention has been paid to research on configuring large-scale centralized energy storage systems within regional grids that host significant penetrations of both utility-scale photovoltaic and wind power generation.

The fundamental principle of power decomposition for HESS involves separating fluctuating power components by frequency, allocating components with different frequencies and amplitudes to distinct forms of energy storage for absorption [14]. Inappropriate allocation not only increases the burden on the storage system but also fails to achieve the complementary advantages of the hybrid system. To address this, researchers have adopted various optimization strategies combined with modal decomposition methods to decompose the total HESS power. Zeyu Ma et al. proposed a double-layer VMD strategy. To mitigate the mode mixing phenomenon in the secondary VMD layer, they incorporated energy entropy thresholds to identify the demarcation point for separating high-frequency and low-frequency components, aiming to achieve a more precise power allocation strategy [15]. Jicheng Fang et al. combined VMD with the Wigner-Ville Distribution algorithm to decompose the net load of an industrial park system [16]. Taiying Zheng et al. introduced a method utilizing VMD to configure the capacity of HESS participating in secondary frequency regulation [17]. Rongchuan Tang et al. proposed a VMD-ST-QF load frequency division strategy, enabling precise decomposition of chronological load curves into high-frequency and low-frequency components [18]. Xinbo Zhou et al. proposed an IBWO-VMD-TCN algorithm that utilizes VMD to decompose historical industrial load into multiple modal components for subsequent prediction [19]. Despite generating relatively effective power allocation strategies, these existing methods exhibit significant subjectivity in selecting key algorithm parameters and overlook the critical step of determining the specific power allocation to different types of energy storage components after the modal decomposition stage. Consequently, these limitations can lead to inaccurate power allocation and excessively high overall costs for the HESS.

Based on the observations and analysis of the aforementioned research, this paper addresses the challenges of collaborative flexibility resource configuration in power grids with high renewable energy penetration by proposing a bi-level optimal planning method for HESS. The main contributions are as follows:

- **A Bi-Level Optimization Model for Multi-Timescale HESS Planning:** Establishes an upper-level planning and lower-level operation collaborative model, integrating HESS siting, sizing, and multi-timescale power allocation. The upper-level focuses on optimizing investment and operation and maintenance (O&M) costs, while the lower-level incorporates flexibility gap penalties, voltage fluctuation, and line loss constraints. This enhances the application scenarios of existing bi-level frameworks for power grids with high penetration of renewable energy, achieving a multi-objective balance among economic efficiency, flexibility, and system stability.
- **An Enhanced VMD Method Integrated with PSO:** Develops PSO-VMD method that adaptively determines key parameter. Eliminates subjectivity in frequency division and improves power allocation accuracy. Introduces median spectrum threshold for precise separation of high/low-frequency components.
- **Comprehensive Validation and Enhanced Algorithm Performance:** Demonstrates superiority via 5 case studies using Southwest China grid data. The developed IWAA integrates Refracted Opposition-based Learning and dynamic crowding distance mechanisms, enhancing global search and preventing premature convergence. Solves complex HESS bi-level planning.

The remainder of this paper is structured as follows. Section 2 delineates the flexibility supply-demand contradiction in high-renewable-penetration power grids. It establishes a hybrid energy storage planning model considering both economic and stability objectives and designs a power co-allocation model for lithium-ion batteries and flow batteries based on the VMD-PSO method. Section 3 proposes the bi-level optimization solution framework utilizing the IWAA algorithm. Section 4 details the modeling and simulation experiments conducted to evaluate the proposed methods and improvements. Finally, Section 5 concludes the paper and discusses potential directions for future research.

2. Methodology

2.1. Modeling

The configuration of HESS can enhance the power supply flexibility of regional grids. If a hybrid energy storage system with sufficiently large power capacity is configured for renewable energy farms, it can functionally balance imbalances in ramp rate, power magnitude, and energy capacity. However, excessive energy storage capacity configuration will lead to substantial unnecessary energy storage installation costs and operational maintenance costs, causing resource waste.

The planning problem for energy storage encompasses economic and stability aspects. The economic aspect refers to the costs throughout the entire lifecycle of energy storage, including investment costs, operational costs, and penalty costs for insufficient system flexibility. The stability aspect includes two dimensions: one is flexibility regulation capability, meaning the stabilizing effect of energy storage on net load fluctuations; the other is the impact of energy storage operation on the distribution network, including voltage fluctuations and line losses. The power system architecture is shown in Figure 1.

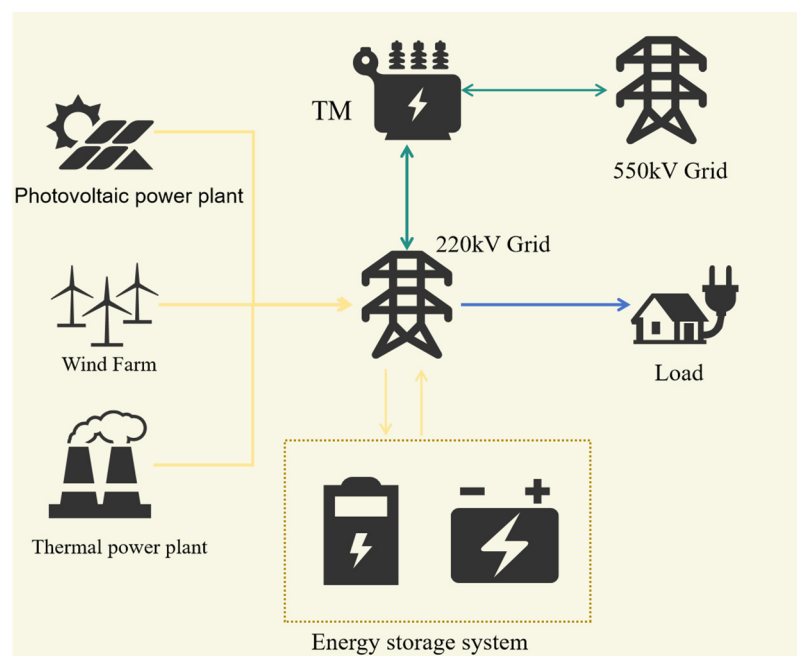


Figure 1. Power grid system.

2.1.1. HESS Model

This chapter selects a flow battery–lithium-ion battery energy storage system to address the system’s flexibility requirements across different time scales. Flow batteries offer long lifespan, rapid response, and adjustable capacity; different storage capacity

demands can be met by adjusting the concentration and volume of the electrolyte. Lithium-ion batteries feature high energy density, response times in the millisecond range, and rapid response to flexibility demands. In this study, the flow battery is configured as the long-duration energy storage and the lithium-ion battery as the short-duration energy storage. Compared to single-type energy storage systems, hybrid energy storage combines complementary advantages: it can frequently charge and discharge to smooth high-frequency fluctuation components, while also possessing large capacity to smooth low-frequency fluctuation components with significant energy magnitude, offering superior economic efficiency and performance. This paper employs an improved VMD algorithm to decompose the power of the energy storage system, generating power signals for the flow battery and lithium-ion battery within the system.

VMD originates from the Empirical Mode Decomposition (EMD) algorithm. Both algorithms can decompose a signal into multiple local oscillatory modes [20]. However, VMD exhibits significant advantages: it is fundamentally grounded in optimization theory, enabling the mathematical description of signals within a variational framework; this ensures unique decomposition results and provides strong adaptability for non-stationary signals. VMD effectively mitigates the common issues in EMD and similar algorithms, such as mode mixing and endpoint effects. It is particularly suitable for processing signals exhibiting significant local frequency variations and demonstrates favorable local characteristics in time-frequency distributions. Consequently, VMD has found widespread application across various domains including signal processing, image processing, and machine learning.

VMD decomposition aims to identify K intrinsic mode functions $u_k(t)$ from non-stationary signals. Each mode $u_k(t)$ possesses a specific bandwidth and a central frequency. The core objective of the solution process is to minimize the sum of the estimated bandwidths of all $u_k(t)$. A fundamental constraint during this optimization process is that the sum of all modal components must equal the original input signal f . Initially, the K obtained $u_k(t)$ modes are processed via the Hilbert transform to derive their respective analytic signals, as shown in Equation (1):

$$(\delta(t) + \frac{j}{\pi t}) * u_k(t), \quad (1)$$

where $*$ is the convolution operator and $\delta(t)$ is the step function.

In order to reduce the error between the reconstructed signal and the original signal, it is necessary to adjust the estimated center frequency of the reconstructed signal and introduce an exponential term to modulate each spectrum into the corresponding fundamental frequency band, as shown in Equation (2):

$$\left[(\delta(t) + \frac{j}{\pi t}) * u_k(t) \right] e^{-j\omega_k t}, \quad (2)$$

The bandwidth of each $u_k(t)$ is estimated by solving the square L_2 norm of the demodulated signal using the above Equation. Thus, the variational problem is constructed as follows:

$$\min_{\{u_k, \omega_k\}} \left\{ \sum_{k=1}^K \int_0^T \left| (\delta(t) + \frac{j}{\pi t}) * u_k(t) \right|^2 e^{-2\omega_k^2 t} dt \right\}, \quad (3)$$

$$s.t. \quad \sum_{k=1}^K u_k(t) = f, \quad (4)$$

The VMD algorithm decomposes the total energy storage target output into K modal components across different frequency bands. The ultimate goal is to identify an appropriate

frequency division point to reconstitute these modal components into high-frequency and low-frequency signal components, with the low-frequency component serving as the target power for lithium-ion batteries and the high-frequency component as the target power for flow batteries. When the number of frequency divisions KL is too high, it leads to excessive decomposition of the energy storage target output. In this case, the sum of the reconstructed components will be less than the actual energy storage target output. Under such circumstances, the actual output of the energy storage system cannot fully mitigate wind power fluctuations, resulting in poor mitigation performance of the energy storage system.

In order to effectively suppress the spectral over-decomposition phenomenon during the decomposition process and to maintain the physical integrity of the output characteristics of the energy storage system, this study proposes an improved energy storage power allocation method incorporating Particle Swarm Optimization (PSO):

Step 1: Optimize the modal component number K and penalty factor α using the PSO algorithm;

Step 2: Set the objective function of the composite loss form:

$$\mathcal{L} = \frac{1}{N} \sum_{t=1}^N \left(x_{\text{orig}}(t) - \sum_{k=1}^K \text{IMF}_k(t) \right)^2 + \lambda \cdot K. \quad (5)$$

where the first term characterizes the full-band reconstruction mean square error (MSE), IMF is the modal component, λ is the complexity penalty factor determined, and the second term K is used to suppress modal redundancy;

Step 3: Solve for the energy center of gravity frequency of each IMF:

$$f_c^{(k)} = \arg \max_{f \in [0, f_s/2]} |\mathcal{F}\{\text{IMF}_k\}|^2. \quad (6)$$

Among them, $\mathcal{F}$ is the Fourier transform operator;

Step 4: Divide the high-frequency component set $\{\text{IMF}_k \mid f_c^{(k)} > \tilde{f}_c\}$ and the low-frequency component set $P_{icl}(t) \{\text{IMF}_k \mid f_c^{(k)} \leq \tilde{f}_c\}$ using the median frequency $\tilde{f}_c$ as the threshold.

Step 5: Output the fused high-frequency group superimposed signal with the reconstructed signal of the low-frequency group containing the residuals.

The VMD-PSO algorithm pseudocode is shown in Algorithm 1.

Algorithm 1. The pseudocode of VMD-PSO.

Algorithm 1: VMD-PSO Hybrid ESS Power Allocation

- 1 Total ESS power P_{ESS} , Mode number K , Penalty factor α , LIB power P_{LIB} , FB power P_{FB}
Initialize PSO particles (K and α combinations);
 - 2 **while** not converged **do**
 - 3 **foreach** particle **do**
 - 4 Perform VMD decomposition: $\min_{\{\omega_k\}} \left\{ \sum_{k=1}^K \left| \left(\delta(t) + \frac{j}{\pi t} \right) * u_k(t) \right| e^{-j\omega_k t/2} \right\};$ //Equation (3)
 - 5 s.t. $\sum_k u_k(t) = P_{ESS}$ //Equation (4)
 - 6 Calculate composite loss $\mathcal{L} = \frac{1}{N} \sum (MSE) + \lambda \cdot K;$ //Equation (5)
 - 7 Compute centroid frequency $f_c^{(k)} = \arg \max |\mathcal{F}\{\text{IMF}_k\}|^2;$ //Equation (6)
 - 8 Split components using median frequency $\tilde{f}_c;$
 - 9 **end**
 - 10 Update global best;
 - 11 **end**
 - 12 Reconstruct high-freq $\rightarrow P_{LIB}$, low-freq $\rightarrow P_{FB};$
-

2.1.2. Upper Level Planning Model

The total economic cost C_{total} incurred over the entire lifecycle of the energy storage system, comprising investment costs and operational costs, is expressed by Equation (7):

$$C_{total} = C_{inv} + C_{OM}, \quad (7)$$

$$C_{inv} = C_E + C_P = u \left(\sum_{k \in \Omega} S_{k,E} E_k + \sum_{k \in \Omega} S_{k,V} P_{k,V} \right), \quad (8)$$

$$\begin{cases} E_k = \sum_{i \in \phi_{ESS}} E_{i,k} \\ P_{k,V} = \sum_{i \in \phi_{ESS}} P_{i,k,V} \end{cases}, \quad (9)$$

$$u = \frac{r(1+r)^{m_k}}{(1+r)^{m_k} - 1}, \quad (10)$$

where C_{inv} denotes the total investment cost, and C_{OM} represents the daily operation and maintenance cost. Equation (8) represents the construction investment cost of the hybrid energy storage system, comprising capacity cost C_E and power cost C_P , $S_{k,E}$ is the unit capacity cost for the k -th energy storage type and $S_{k,V}$ is the unit power cost for the k -th energy storage type. Equation (9) defines the rated power and rated capacity of energy storage construction, where $k \in \{1: \text{Lithium Battery}, 2: \text{Flow Battery}\}$ and ϕ_{ESS} denote the sets of integrated energy storage units. In Equation (10), u represents the present value, indicating the discounted current worth of an investment amount after m_k years at an annual interest rate r .

The operation and maintenance (O&M) cost of an energy storage plant refers to the expenses required to keep the storage system in good standby condition. This cost includes routine maintenance expenses as well as the expected annualized cost needed to maintain the system's power and capacity without degradation over its entire lifecycle. The calculation is as follows:

$$C_{OM} = \sum_{k \in \Omega} (c_{OM,k}^P \sum_{t=1}^T |P_{k,ESS}(t)|) + \sum_{k \in \Omega} (c_{OM,k}^Q E_k), \quad (11)$$

where $c_{OM,k}^P$ and $c_{OM,k}^Q$ represent the annual average power maintenance cost coefficient and annual average capacity maintenance cost coefficient of the k th type of energy storage, respectively. T represents the number of samples, and $P_{k,ESS}(t)$ represents the energy storage operating power of energy storage k at time t .

2.1.3. Lower Level Operational Mode

To verify the impact of the energy storage capacity and power obtained from the upper-level planning on the stability of new-type power systems, we constructed a lower-level multi-objective optimization model aiming to enhance system flexibility and ensure grid operational stability. The Equation is detailed as follows:

$$\min f = \min[f_1, f_2, f_3, f_4], \quad (12)$$

where F is the objective function for lower-level energy storage operation optimization, f_1 represents operational cost, f_2 denotes the gap between the actual operation power of energy storage and the system flexibility deficiency penalty, f_3 indicates voltage deviation, and f_4 signifies line loss.

Based on the research by Chen et al. [21], O&M costs incurred during energy storage operation share the same timescale with voltage fluctuations, line losses, and flexibility deficiency penalties. Therefore, these O&M costs are incorporated into the operational optimization model, with the cost Equation given in Equation (13).

$$f_1 = C_{OM} , \quad (13)$$

$$f_2 = 365 \cdot \sum_{s \in S_n} (p_s C_{FSD,s}) , \quad (14)$$

$$C_{FSD} = Q_{penal} \sum_{t=1}^T P_{FSD}(t) , \quad (15)$$

$$P_{FSD}(t) = |\theta(t) - P_{ESS}(t)| , \quad (16)$$

$$\theta(t) = P_{load}(t) - P_{grid}(t) - (P_{PV}(t) + P_{WD}(t) - P_{aban}(t)) - P_{TPP}(t) , \quad (17)$$

$$P_{ESS}(t) = \sum_{i \in \phi_{ESS}} P_{i,ESS}(t) . \quad (18)$$

Flexibility supply-demand equilibrium means that the system has more than the permitted level of abundance of flexibility supply relative to flexibility demand for each type of resource at any moment, under any time scale, and in any direction. When the model in this chapter configures energy storage, there is a situation where the storage output power does not fully cover the system flexibility power deficit. In order to enhance the system flexibility regulation capability and reduce the loss of energy storage due to insufficient capacity and power allocation, this paper sets the system flexibility insufficiency penalty cost to limit the situation in which the energy storage output power cannot cover the system flexibility power shortage.

Equation (14) represents the annual flexibility penalty cost of the power system operation. S_n represents the set of several typical scenarios for the joint output of wind power, photovoltaic power, and load. p_s represents the probability of occurrence for each scenario. Equation (15) represents the penalty cost caused by insufficient system flexibility supply under scenario S . For ease of expression, the cost is calculated for different scenarios, as shown in Equations (9)–(12). Q_{penal} represents the unit penalty cost for insufficient system flexibility supply. Equation (16) represents the difference between the energy storage operating power and the total system flexibility power shortage at time t , where $P_{ESS}(t)$ is the total energy storage operating power at time t . Equation (17) is the grid flexibility gap Equation. This paper considers the output of renewable energy, thermal power output, and energy storage output as the flexibility supply resources of the power system, with the load demand being the flexibility demand. Here, P_{load} is the load demand, P_{grid} is the power exchange with the main grid, P_{PV} is the photovoltaic generation power, P_{WD} is the wind generation power, P_{aban} is the wind and photovoltaic curtailment power, and P_{TPP} is the thermal power generation. The energy storage operating power is composed of the sum of the operating powers of all configured energy storage nodes ($i \in \phi_{ESS}$), as shown in Equation (18), where $P_{i,ESS}(t)$ is the energy storage operating power at node i at time t .

$$f_3 = \sum_{s \in S_n} (p_s \Delta U_s) , \quad (19)$$

$$\Delta U = \sum_{t=1}^T \sum_{i \in \phi}^N |U_i(t) - \bar{U}(t)|, \quad (20)$$

$$\bar{U}(t) = \frac{\sum_{i \in \phi}^N U_i(t)}{N}. \quad (21)$$

Equation (19) is the power system voltage fluctuation Equation (20) represents the system voltage fluctuation level equation, where U_i is the voltage at node i at time t , and n is the number of nodes. Equation (21) calculates the average voltage $\bar{U}(t)$ across all nodes in the system at time t .

$$f_4 = \sum_{i=1}^T \sum_{j=1}^N R_{i,j} \frac{P_{i,j}^2 + Q_{i,j}^2}{U_{i,j}^2}, \quad (22)$$

Equation (22) is the system line loss equation, where $R_{i,j}$ is the resistance of the branch between nodes i and j . $P_{i,j}$ and $Q_{i,j}$ are the active power and reactive power of the branch between nodes i and j , respectively.

2.1.4. Objective Function

The objective function is shown in Equation (23). Configuring energy storage can promote the accommodation of wind and photovoltaic power and reduce wind and photovoltaic curtailment, but may simultaneously be accompanied by an increase in investment costs. Therefore, capacity configuration requires multi-dimensional consideration. This chapter comprehensively considers the economy, flexibility supply capability, and power supply reliability of the hybrid energy storage system to establish a multi-objective function, expressed as:

$$\min F = \min \sum (C_{total}, f(f_1, f_2, f_3, f_4)). \quad (23)$$

2.1.5. Constraints

Considering the impact of centralized energy storage configuration on grid stability, and combining the energy storage's own charge–discharge balance constraint and operating power constraint, the following constraints are obtained:

Excessive charging and discharging of the energy storage system will both lead to a substantial reduction in its cycle life. Different types of energy storage have different optimal capacity operating intervals. Limiting the operating interval of the energy storage capacity within the optimal interval can effectively increase the number of cycles and service life of the battery. This paper focuses solely on capacity constraints for research, ignoring the energy attenuation of the energy storage during use.

$$SOC_{i,\min} \leq SOC_i(n) \leq SOC_{i,\max}, \quad (24)$$

where $SOC_{i,\min}$ and $SOC_{i,\max}$ represent the upper and lower limits of the remaining power of the i -th type of energy storage, respectively.

The actual charging and discharging power of the energy storage at each moment during operation is less than the rated power configured at each node. As shown in Equation (25):

$$\begin{cases} E_i^{\text{ESS},\min} \leq E_i^{\text{ESS}} \leq E_i^{\text{ESS},\max} \\ P_i^{\text{ESS},\min} \leq P_i^{\text{ESS}} \leq P_i^{\text{ESS},\max} \end{cases} \quad (25)$$

where $E_i^{\text{ESS,min}}$ and $E_i^{\text{ESS,max}}$ denote the upper and lower limits of the rated capacity of the energy storage installed at node i ; $P_i^{\text{ESS,min}}$ and $P_i^{\text{ESS,max}}$ denote the upper and lower limits of the rated power of the energy storage installed at node i , respectively.

$$\sum_{r \in R_k} X_r^{\text{cha}} + X_r^{\text{dis}} = 1, \quad (26)$$

Energy storage devices of the same type at moment t cannot be charged or discharged simultaneously. Equation (26) represents the state of the energy storage device at moment t , where X_r^{cha} and X_r^{dis} , respectively, represent the charging and discharging state of the energy storage device at time t , which are 0–1 variables, and $R_k, k \in \{1: \text{Lithium Battery}, 2: \text{Vanadium Flow Battery}\}$ represents the type of energy storage.

$$P_{TH}(t) + P_{\text{wind/PV}}(t) - P_{\text{aban}}(t) + P_{\text{grid}}(t) = P_{\text{load}}(t) + P_{\text{ESS}}(t) - P_{\text{cl}}(t), \quad (27)$$

Equation (27) represents the flexibility supply-demand power balance of the power grid system at time t . Here, $P_{TH}(t)$ represents the thermal power at time t , $P_{\text{wind/PV}}(t)$ represents the joint output power of wind and photovoltaic at time t , $P_{\text{load}}(t)$ represents the total load power of the regional power grid at time t , $P_{\text{ESS}}(t)$ represents the charging and discharging power of the hybrid energy storage system at time t , $P_{\text{aban}}(t)$ represents the wind and photovoltaic curtailment power of the grid at time t , and $P_{\text{cl}}(t)$ represents the load shedding power at time t . $P_{\text{grid}}(t)$ represents the tie-line power between the regional power grid and the main grid at time t , which is divided into downward power $P_{\text{grid,Down}}(t)$ and upward power $P_{\text{grid,UP}}(t)$; here, downward power represents the power purchased by the grid system from the main grid, and upward power represents the power sold to the main grid.

$$\begin{cases} P_{\text{grid,DOWN}}(t) = P_{\text{grid}}(t) & P_{\text{grid}}(t) \geq 0 \\ P_{\text{grid,UP}}(t) = -P_{\text{grid}}(t) & P_{\text{grid}}(t) < 0 \end{cases}, \quad (28)$$

$$P_{\text{grid}}(t) = P_{\text{grid,DOWN}}(t) - P_{\text{grid,UP}}(t) = P_{\text{purc}}(t) - P_{\text{sell}}(t), \quad (29)$$

$$\begin{cases} U_{i,\min} \leq U_{i,j} \leq U_{i,\max} \\ I_{ij,\min} \leq I_{ij,j} \leq I_{ij,\max} \end{cases}, \quad (30)$$

Equation (30) represents the grid operation security constraint. Where $U_{i,\max}$ and $U_{i,\min}$ are the upper and lower limits of the voltage at node i , $I_{ij,\min}$ and $I_{ij,\max}$ are the minimum and maximum values of the allowable branch current, respectively.

$$\begin{cases} P_i(t) + \sum_{k \in \phi_i} P_{ik}(t) - \sum_{j \in \phi_i} (P_{ij}(t) - r_{ij} I_{ij}(t)^2) = 0 \\ Q_i(t) + \sum_{k \in \phi_i} Q_{ik}(t) - \sum_{j \in \phi_i} (Q_{ij}(t) - x_{ij} I_{ij}(t)^2) = 0 \end{cases} \quad (31)$$

$$I_{ij}(t)^2 = \frac{P_{ij}(t)^2 + Q_{ij}(t)^2}{U_i(t)^2}, \quad (32)$$

$$U_j(t)^2 = U_i(t)^2 - 2(P_{ij}(t)r_{ij} + Q_{ij}(t)x_{ij}) + I_{ij}(t)^2(r_{ij}^2 + x_{ij}^2), \quad (33)$$

Equation (31) represents the active and reactive power balance of a node. Where ϕ_c is the set of head nodes of all branches whose end nodes are i . ϕ_a is the set of head nodes of all branches whose head nodes are i , $P_i(t)$ and $Q_i(t)$ are the active and reactive loads

of node i at time t , respectively. U_i is the voltage of node i . $P_{ij}(t)$ and $Q_{ij}(t)$ are the active and reactive power of branch ij at time t , respectively.

Equation (32) represents the branch current constraints in the power system, where r_{ij} and x_{ij} are the resistance and reactance of branch ij , respectively. In this paper, each branch is numbered according to the node number at its end point, and the branch power flows from the node with the smaller number to the node with the larger number in the positive direction. Equation (33) represents the node voltage balance constraint.

$$P_i(t) = P_{i,load}(t) - P_{i,WD/PV}(t) - P_{i,ESS}(t) + P_{i,aban}(t) - P_{i,grid}(t) - v_i(t)P_{i,cl}(t) - P_{i,TPP}(t), \quad (34)$$

$$Q_i(t) = Q_{i,load}(t) - Q_{i,WD/PV}(t) - Q_{i,grid}(t) - v_i(t)Q_{i,cl}(t) - Q_{i,TPP}(t), \quad (35)$$

Equations (34) and (35) represent the active and reactive power balance constraints for nodes, where $P_{i,load}(t)$, $P_{i,WD/PV}(t)$, $P_{i,ESS}(t)$, $P_{i,aban}(t)$, $P_{i,grid}(t)$, $P_{i,cl}(t)$, and $P_{i,TPP}(t)$ denote the active power load at node i at time t , the active power output from wind and solar power plants, the charging and discharging power of energy storage systems, wind and solar power curtailment power, active power injected from the upper-level main grid, active power from interruptible load, and thermal power plant active power. If a node does not have these resources connected, the corresponding variables are set to 0. $Q_{i,load}(t)$, $Q_{i,WD/PV}(t)$, $Q_{i,grid}(t)$, $v_i(t)Q_{i,cl}(t)$, and $Q_{i,TPP}(t)$ represent the reactive power of load, wind and solar power generation, upper-level grid injection, interruptible load, thermal power plant, and hydroelectric power at node i at time t .

3. Problem Description and Modeling Double-Layer Optimization Configuration Solution Process

This study establishes a mixed-integer nonlinear bi-level optimization model for hybrid energy storage systems. The upper layer determines storage location, rated capacity, and rated power to minimize annualized lifecycle costs. The lower layer allocates storage power to optimize flexibility regulation and voltage stability. Power signals are decomposed into high-frequency (lithium-ion batteries) and low-frequency (flow batteries) components using frequency separation. These results feed back to the upper layer for iterative refinement of capacity/power planning, creating a closed-loop between long-term planning and short-term operation. The Weighted Average Algorithm (WAA) solves both optimization layers. The solution process for hybrid energy storage double-layer optimization is shown in Figure 2.

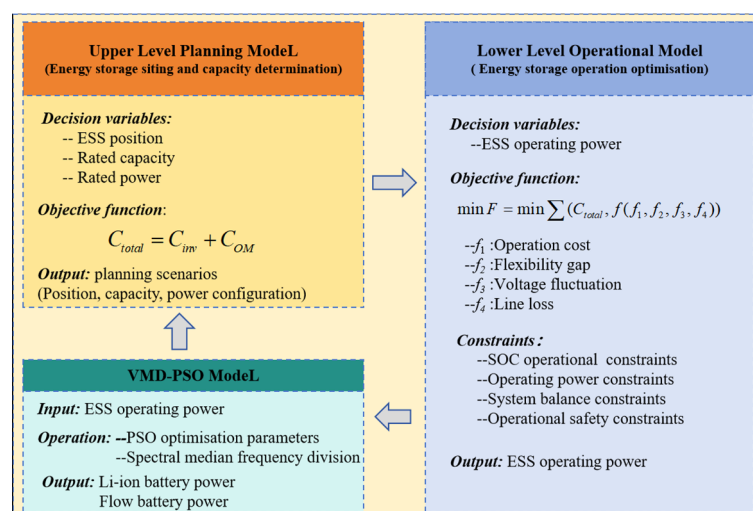


Figure 2. HESS planning process.

3.1. Weighted Average Algorithm

In this paper, the upper level planning is a mixed integer nonlinear problem and the lower run is a non-convex optimization, which cannot be solved directly by traditional mathematical planning methods such as KKT conditions, and heuristic algorithms are commonly used for energy storage planning problems of similar complexity [22]. Weighted average algorithm (WAA) is a novel metaheuristic algorithm inspired by the concept of weighted average position. Unlike previous swarm intelligence algorithms, this algorithm requires constructing the weighted average position of the entire population during each iteration, preventing individuals in the population from becoming trapped in local optima [23].

The main idea of the WAA is to apply the weighted average position to search for near-optimal solutions. When calculating the overall average position, it is necessary to consider the weights of each element in the dataset; elements with larger weights exert a more significant influence on the final average position.

3.1.1. Population Initialization

The optimization process first starts from a $D \times N$ dimensional space, where N is the population size, D represents the space dimension, and the position of each individual in the population is defined as X . Equation (36) represents that the individual position vector X is composed of N populations each of dimension d :

$$X = rand(N, D) \times (ub_n^i - lb_n^i) + lb_n^i, \quad (36)$$

$$n \in [1, N], i \in [1, D], \quad (37)$$

where ub_n^i is the upper search bound of the n -th population in the i -th dimension, and lb_n^i is the lower search bound of the n -th population in the i -th dimension.

3.1.2. Weighted Average Position

In the WAA, it is first necessary to calculate the fitness of each participant and rearrange the population based on one of two quality characteristics—that is, whether a larger individual indicates superior population fitness or a smaller one indicates superior fitness—thereby rearranging the population. Then, the top $N_{\text{Candidate}}$ individuals are selected from the population.

$$N_{\text{Candidate}} = (nP - 4) \frac{it - 1}{\text{Max}_{it}} + nP, \quad (38)$$

$$\text{Sum}_{\text{Fitness}} = \sum_{i=1}^{N_{\text{Candidate}}} \text{Fitness}(X_i), \quad (39)$$

$$X_{\text{Miu}} = \frac{\sum_{i=1}^{N_{\text{Candidate}}} X_i (\text{Sum}_{\text{Fitness}} - \text{Fitness}(X_i))}{\text{Sum}_{\text{Fitness}} (N_{\text{Candidate}} - 1)}, \quad (40)$$

$$X_{\text{Miu}} = \sum_{i=1}^{N_{\text{Candidate}}} X_i \cdot \text{Fitness}(X_i) / \text{Sum}_{\text{Fitness}}, \quad (41)$$

Equation (38) represents the number of individuals selected from the population, where nP is the population size, it is the current iteration number, and Max_{it} is the maximum iteration number. $\text{Sum}_{\text{Fitness}}$ is the sum of all fitness values of the selected candidates. Equations (39) and (40) are the calculation equations for the high-quality region center of the current population. Equations (39) and (40) represent the update methods of the weighted average position in minimization problems and maximization

problems, respectively. $\min$ represents the quality center of the current population, guiding the subsequent movement direction.

3.1.3. Exploration and Exploitation Switching Mechanism

During the iteration process, Equation (42) is applied to determine the search phase for candidate solutions, namely exploration or exploitation:

$$f(it) = (\alpha \cdot rand - 1) \sin\left(\pi \frac{it}{Max_{it}}\right), \quad (42)$$

where it is the current iteration number, Max_{it} is the maximum iteration number, α is a constant used to control the balance between exploration and exploitation phases. In this paper, 0.5 is set as the threshold to achieve balance between exploitation and exploration. More specifically, candidate solutions with $f(it) \geq 0.5$ will move according to exploitation capability. Under conditions below the threshold, candidate solutions will move according to exploration capability. It is conceived that mutual cooperation between exploration and exploitation will yield a better search space.

The first strategy focuses on utilizing the search space, which is determined by the weighted average position of the entire current population as well as the individual best position and the global best position, as shown below:

$$X_i(it+1) = w_{11} \cdot (X_{Mit}(it) - X_{Global_{best}}(it)) + w_{12} \cdot (X_{Mit}(it) - X_{Personal_{best}}(it)) + w_{13} \cdot X_{Mit}(it), \quad (43)$$

where w_{11} , w_{12} , and w_{13} are random numbers between 0 and 1, used to adjust the searchable space of the individual best position $X_{Personal_{best}}$ and the global best position $X_{Global_{best}}$.

The second approach concentrates on utilizing the region defined by the weighted mean location of all individuals and each particle's own optimal position. Unlike the first approach, this method disregards the global optimum, thereby reducing the extent of the search area. The corresponding search region is formulated in Equation (44).

$$X_i(it+1) = w_{21} \cdot (X_{Mit}(it) - X_{Personal_{best}}(it)) + w_{22} \cdot X_{Personal_{best}}(it), \quad (44)$$

where w_{21} and w_{22} are random values between 0~1. Compared to the first strategy, this movement strategy is more inclined to enhance convergence speed and accuracy.

The third approach targets the search region bounded by the global optimal position and the population's weighted mean position. In this method, each candidate's new position is determined according to the following calculation:

$$X_i(it+1) = w_{31} \cdot (X_{Mit}(it) - X_{Global_{best}}(it)) + w_{32} \cdot X_{Global_{best}}(it), \quad (45)$$

where w_{31} and w_{32} are random values between 0~1. The search space of this strategy will be smaller than that of the first and second strategies. Movement from the global best position to the weighted average position will enhance convergence speed and accuracy.

3.1.4. Exploration Phase

In the WAA algorithm, two strategies are employed to balance the relationship between global and local search during the optimization process within the feasible solution space, enabling the algorithm to identify the optimal solution.

Lévy flight is a random walk model conforming to the Lévy distribution, exhibiting Markov properties during its process. The step sizes simulated in the Lévy flight model are predominantly small but occasionally exhibit large steps, as shown in Equation (48).

$$S = \frac{U}{|V|^{1/\beta}}, \quad (46)$$

$$U, V \sim N(0, \sigma_u^2), \quad (47)$$

$$\sigma_u = \left\{ \frac{\Gamma(1+\beta) \sin(\pi\beta/2)}{\Gamma\left[\frac{(1+\beta)}{2}\right] \beta 2^{(\beta-1)/2}} \right\}^{1/\beta}, \quad (48)$$

Equation (46) represents the step size of Lévy's flight, where β is the step control parameter that takes into account the long and short distance movement of the individual.

Based on the controlled adjustment of the step length S , the first exploration strategy is given by the following Equation (49):

$$X_{i,j}(it+1) = X_{Global_{best,j}}(it) + S, \quad (49)$$

where S represents the step size of the Lévy flight, and $X_{Global_{best,j}}(it)$ denotes the global optimal solution at position j during the it -th iteration. $X_{i,j}(it+1)$ indicates the j -th position of the i -th solution after $(it+1)$ iterations.

To enable the population to escape local optima when following the first strategy, the WAA applies a second movement strategy to adjust the search space:

$$X_{i,j}(it+1) = rand \cdot (ub_{min} - lb_{min}) + LB_{min}, \quad (50)$$

where lb_{min} and ub_{min} are defined as the minimum values of the lower and upper bounds for each dimension. Through this movement strategy, the search agent will move to a new position to seek other better regions.

3.2. Improvement Strategy

The standard WAA provides a robust foundation for optimization but exhibits two primary limitations when solving the complex, high-dimensional bi-level HESS planning problem formulated in this study:

In preliminary tests, the standard WAA occasionally converged to local optima, particularly when initial populations were distributed in less favorable regions of the search space. This is attributed to its exploitation-dominated search strategy in later stages, which lacks an effective mechanism to escape local basins of attraction.

For multi-objective optimization, maintaining a diverse and well-distributed set of non-dominated solutions is critical. The conventional crowding distance calculation, used for niching and selection, often fails to preserve solution diversity uniformly across the objective space, especially with more than two objectives. To address these challenges, the Improved WAA (IWAA) incorporates two targeted strategies.

3.2.1. Reflective Reverse Learning Strategy

This paper introduces the refraction opposition-based learning strategy into the second optimization approach of the WAA exploration phase. The refraction opposition-based learning mechanism is an excellent improvement mechanism for optimizing initial assignments and candidate solutions. Building upon opposition-based learning, it combines the law of light refraction to search for the optimal solution, enabling the algorithm to jump out of the current optimization position region and direction during the WAA exploration phase. The Equation is as follows:

$$X_{i,j}(it+1) = \begin{cases} \frac{ub_{i,j} + lb_{i,j}}{2} + \frac{ub_{i,j} + lb_{i,j}}{2 \cdot it} - \frac{X_{i,j}(it)}{it} & r \leq 0.5 \\ \frac{ub_{i,j} + lb_{i,j}}{2} + \frac{ub_{i,j} + lb_{i,j}}{2 \cdot it} - \frac{X_{i,j}(it)}{it} & 0.5 \leq r < 1 \end{cases}, \quad (51)$$

Among them, $X_i(it+1)$ is the inverse reflection solution of $X_i(it)$, and r is a random value between 0 and 1.

3.2.2. Dynamic Dense Distance Based Noninferior Solution Set Updating

During multi-objective optimization, the non-dominated solution set must be updated after each iteration. To maintain the size of the Pareto solution set and ensure uniformity in solution distribution, it is necessary to perform “superior solution selection” on Pareto solutions. The crowding distance characterizes the density between a particle and its neighboring particles, describing the uniformity of solutions. For an optimization problem with three objective functions f_1 , f_2 and f_3 , the crowding distance $I(x_i)$ of particle x_i is:

$$I(x_i) = \sum_{m=1}^n \left[|f_m(x_j) - f_m(x_k)| / f_{m\max} \right]. \quad (52)$$

After calculating the crowding distance for each Pareto solution, sort them in descending order of crowding distance and perform screening. The conventional method involves selecting the top N solutions with the largest crowding distance in sequence. Although computationally efficient—requiring only one crowding distance calculation per iteration—this approach often compromises the diversity and uniformity of the selected Pareto solutions. This paper adopts the “sequential removal” method for updating non-dominated solutions: after sorting by crowding distance, remove the solution with the smallest crowding distance, recalculate the crowding distance for the remaining Pareto solutions, resort them, and again remove the solution with the smallest crowding distance. This process is iteratively repeated until the number of remaining Pareto solutions reaches N . The improved ETO algorithm pseudocode and the overall model solution method pseudocode are shown in Algorithm 2 and Algorithm 3, respectively.

Algorithm 2. The pseudocode of IWAA.

Algorithm 2: Improved Weighted Average Algorithm (IWAA)

Input: Population size N , Dimension D , $MaxIter$

Output: Optimal solution X_{best}

1 Initialize positions $X = rand(N, D) \times (ub_n^i - lb_n^i) + lb_n^i$; //Equation (36)

2 for $t \leftarrow 1$ to $MaxIter$ do

3 **Calculate quality center** X_μ : **Select best** $N_{candidate}$ **solutions**; //Equation (38)

4 Compute $X_\mu = \frac{\sum X_i (SumFitness - Fitness(X_i))}{SumFitness(N_{candidate} - 1)}$ //Equation (40)

 //Phase switching

5 **if** $f(it) = (\alpha \cdot rand - 1) \sin(\pi \cdot it / Max_{It}) \geq 0.5$; //Equation (42)

6 **then**

 //Exploitation strategies

7 **Strategy 1:** $X_i(it+1) = w_{11} \cdot (X_u - X_{Global_{best}}) + w_{12} \cdot (X_u - X_{Personal_{best}}) + w_{13} \cdot X_u$ //Equation

(43)

8 **Strategy 2:** $X_i(it+1) = w_{21} \cdot (X_u - X_{Personal_{best}}) + w_{22} \cdot X_{Personal_{best}}$ //Equation (44)

9 **Strategy 3:** $X_i(it+1) = w_{31} \cdot (X_u - X_{Global_{best}}) + w_{32} \cdot X_{Global_{best}}$ //Equation (45)

10 **else**

```

//Exploitation strategies
11   Refracted opposition:  $X_i^{t+1} = \frac{ub+lb}{2} + \frac{ub+lb}{2t} - \frac{X_i}{t}$ ; //Equation (51)
12   Levy flight:  $X_i^{t+1} = X_{Global} + Levy(\beta)$  //Equations (46)–(49)
13   end
//Pareto set maintenance
14   Calculate crowding distance  $I(x_i) = \sum_{m=1}^n [ |f_m(x_j) - f_m(x_k)| / f_{mmax} ]$ ; //Equation
(52)
15   Sort by descending  $I(x_i)$ ;
16   Remove solutions with smallest crowding distance until  $|ParetoSet| \leq N$ ;
17 end
18 return  $X_{best}$ 

```

Algorithm 3. The pseudocode of HESS Bi-level Optimization framework.

Algorithm 3: HESS Bi-level Optimization Framework

Input: Wind power P_W , PV power P_{PV} , Load demand P_L , Candidate nodes

Output: Optimal ESS configuration {location, capacity, power}, Pareto frontier

1 Initialize parameters: $MaxIter$, Population size N , ESS types $k \in \{LIB, FB\}$;

2 Initialize population: Random ESS configurations (location, capacity, power);

3 for $t \leftarrow 1$ to $MaxIter$ do

 //Upper-level: Ess planning

4 foreach configuration X_i do

5 **Calculate lifecycle cost** $C_{total} = C_{inv} + C_{OM}$ //Equations (7)–(11)

 //Lower-level: operational optimization

6 Calculate flexibility gap: $\theta(t) = P_L - P_g - (P_{PV} + P_W - P_{aban}) - P_{TPP}$; //Equation (17)

7 Execute **Algorithm 1 (VMD-PSO)** for power allocation;

8 Calculate penalty cost $C_{FSD} = Q_{penal} \sum |\theta(t) - P_{ESS}(t)|$; //Equation (15)

9 Calculate voltage deviation ΔU and line losses f_4 ; //Equations (19)–(22)

10 Return objectives: $F = [C_{total}, C_{FSD}, \Delta U, f_4]$ //Equation (23)

11 end

12 Execute **Algorithm 2 (IWAA)** to update configurations;

13 Update Pareto set using dynamic crowding distance;

14 end

15 return optimal FSS configuration and Pareto frontier

4. Simulation and Analysis

In order to verify the validity of the model and methodology proposed in this paper, the problem of hybrid energy storage siting and capacity determination is investigated in the power grid of a region with high penetration rate of new energy. This paper selects the new energy output and load data of a place in Southwest China as the research object, the power supply area of the regional power grid is 12,735 square kilometers, covering a power supply population of 5.68 million people, installed capacity of thermal power units is 3500 MW, and serving 2.33466 million customers. 2023, the regional power grid completed the annual sales of 31.41 billion kWh, with the highest daily power supply of 106 million kWh, and the peak and valley load difference of 3091 MW, which is a

significant pressure on the grid peak regulation. The pressure of grid peaking is significant. The clustering results of the annual output curves of wind, light and load in the system are shown in Figures 3–5, and the coverage of the colored area near the solid line represents the 95% confidence interval of this clustering scenario.

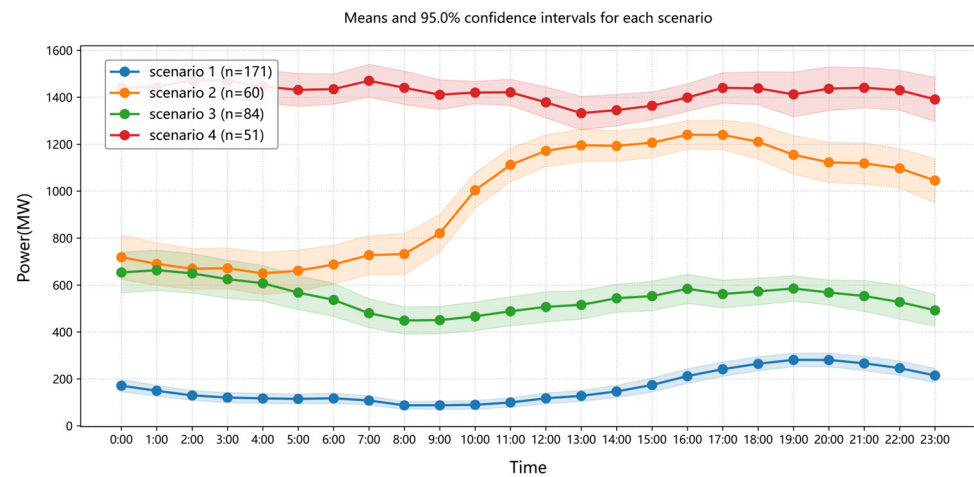


Figure 3. The clustering result of wind power.

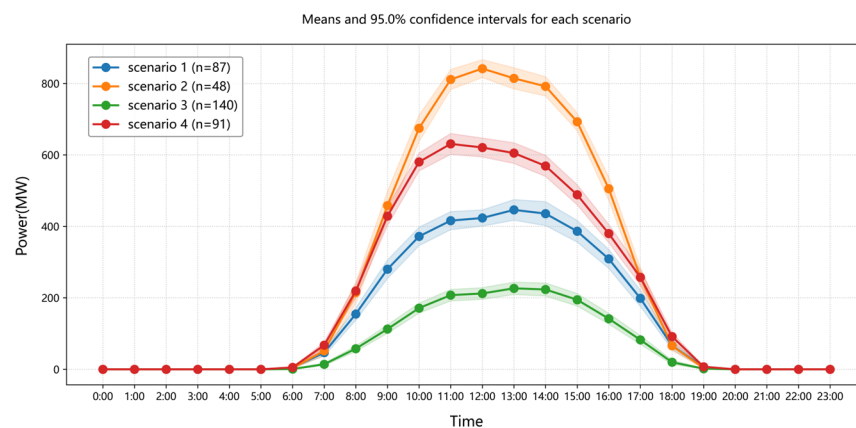


Figure 4. The clustering result of PV power.

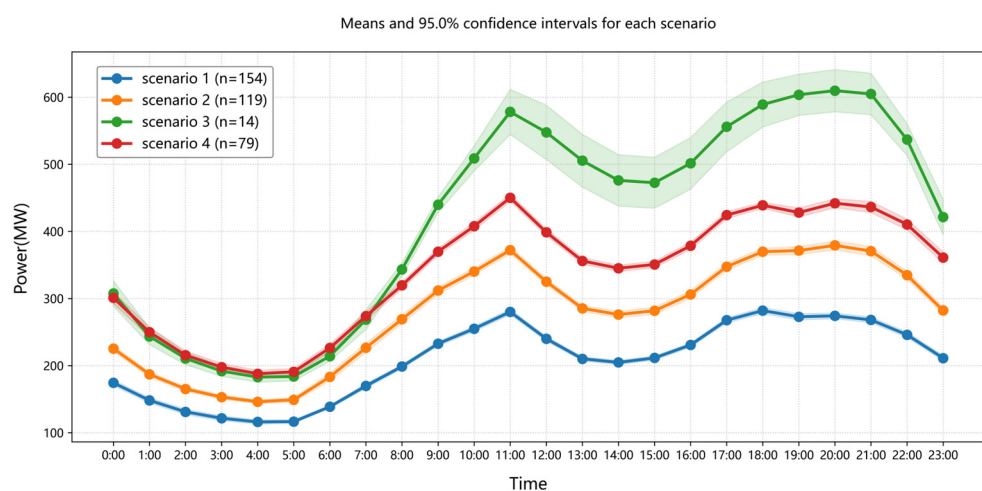


Figure 5. The clustering result of Load power.

It can be observed that the fluctuation of clustering centers for historical wind power, PV, and load data across the respective regions shows significant disparities. The

clustering results provide data input for subsequent planning tasks. This paper verifies the effectiveness of the proposed method using a modified IEEE 39-bus transmission network. The grid voltage level is 220 kV, and node 1 is the reference bus, which connects to the upstream grid through a substation. Thermal power units are installed at nodes 31 and 33; photovoltaic units are installed at node 30; wind power units are installed at nodes 36, 37, and 38. The simulation is conducted using real operational data obtained from the power grid company of the case city. MATLAB R2024b serves as the simulation platform, with computational environment settings as follows: Windows 10 operating system, 16 GB RAM, and an Intel Core i7-11800H CPU. Energy storage parameter configurations are listed in Table 1 [24].

This paper establishes five different planning schemes: Scheme 1 deploys no energy storage; Scheme 2 deploys only lithium-ion batteries; Scheme 3 deploys only flow batteries; Scheme 4 deploys hybrid energy storage at the same node; Scheme 5 represents the proposed optimization scheme. Comparative results are shown in Table 2. In this Table, the installation location refers to the bus number in the 39-bus system.

Table 1. Energy storage parameter settings.

	Lithium Ion Battery	Flow Battery
Unit capacity cost (CNY/KWh)	3000	5000
Unit power cost (CNY/KW)	9000	3600
Present value of funds discount factor	6%	
Expected operational life of energy storage (years)	20	30
Average annual maintenance cost factor for energy storage k (CNY/Wh)	0.06	0.065
System inflexibility unit penalty cost (CNY/KWh)	1.5	
Energy storage SOC range (%)	[20, 80]	[10, 90]

Table 2. Comparison of planning result of different schemes.

Scheme	Installation Location	Lithium-Ion Battery Capacity (MWh)	Flow Battery Capacity (MWh)	Lithium-Ion Battery Power (MW)	Flow Battery Power (MW)	Voltage Fluctuation (pu)	Line Loss (pu)	Penalization Cost (CNY)	Total Cost (CNY)
Scheme 1	-	-	-	-	-	56.55	106.94	1.6042×10^7	1.6042×10^7
Scheme 2	33	5370	-	3280	-	26.15	58.12	4.0281×10^6	1.1638×10^7
Scheme 3	21	-	5700	-	4400	28.71	60.74	5.3612×10^6	1.0736×10^7
Scheme 4	15	2410	4860	550	3880	24.05	50.63	2.9062×10^6	9.1448×10^6
Scheme 5	21, 15	2600	5300	2160	4500	20.52	45.27	2.3960×10^6	8.7718×10^6

According to the comparative analysis of Table 2, when no energy storage is configured in the system, reliance solely on main grid transmission and thermal power peak regulation to address renewable energy fluctuations results in voltage fluctuations as high as 56.55 p.u., line losses of 106.94 p.u., and penalty costs of 1.6042×10^7 due to severe flexibility deficiency, with total costs far exceeding other solutions. For Scheme 2 employing single lithium-ion battery storage, although voltage fluctuations are reduced by 53.8% and line losses decreased by 45.7% through rapid power response, the penalty cost still reaches 4.0281×10^6 due to limited capacity regulation capability, exposing the shortcoming of short-duration storage in addressing sustained energy deficits.

Scheme 3 with single flow battery solution, constrained by insufficient power response speed, achieves 49.2% reduction in voltage fluctuations and 43.2% decrease in line losses compared to Scheme 1. However, it exhibits higher losses than Scheme 2, and due to long-duration regulation requirements, its penalty cost exceeds that of the lithium-based solution. Its economic viability is also compromised by high investment costs.

Scheme 4 adopts hybrid energy storage at the same node. Through lithium-ion and flow battery coordination, voltage fluctuations and line losses are significantly optimized, and penalty costs plummet to 2.9062×10^6 , validating the operational logic where high-frequency fluctuations are smoothed by lithium-ion batteries and low-frequency energy is regulated by flow batteries. However, its co-located centralized configuration results in residual power/capacity redundancy, with insufficiently significant total cost improvement compared to Schemes 2 and 3.

After implementing the optimized solution proposed in this paper, multi-node coordinated deployment effectively integrates the characteristics of load and renewable energy integration in the grid system. Total cost is reduced to 8.7718×10^6 , representing a 4.1% decrease compared to the sub-optimal Scheme 4. Moreover, voltage fluctuations, line losses, and penalty costs are significantly reduced relative to other schemes, enhancing system economy and stability.

To validate the effectiveness of the proposed improved VMD algorithm for hybrid energy storage control, this paper takes Scheme 5 as an example, with lithium-ion battery power at 2160 MW and flow battery power at 4500 MW. First, the proposed method decomposes the energy storage system power into four components. IMF1~IMF4 are the modal components of the total storage system power, showing progressively decreasing fluctuation amplitudes with increasing modal order. Subsequently, high-frequency and low-frequency components are partitioned, assigning the high-frequency components to lithium-ion battery operating power and the low-frequency components to flow battery power. The reconstruction results between IMFs are shown in Figure 6 and the power curves of the two batteries are shown in Figure 7.

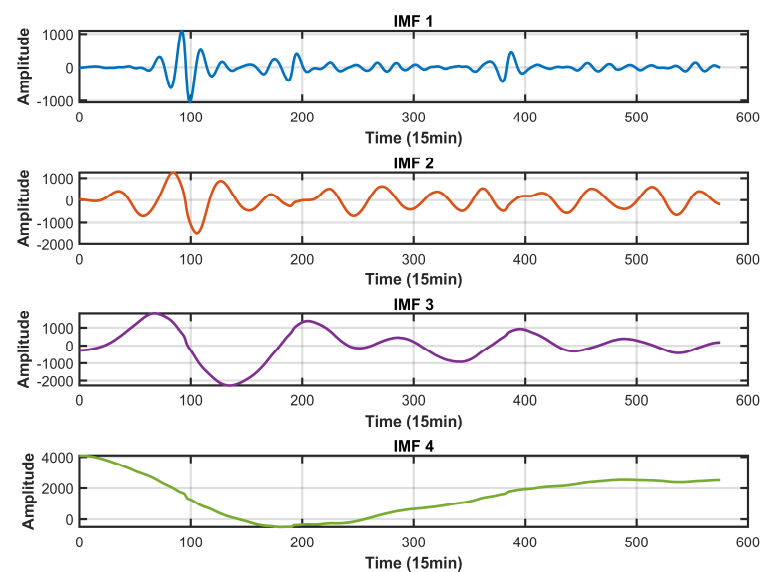


Figure 6. The Results for each IMF.

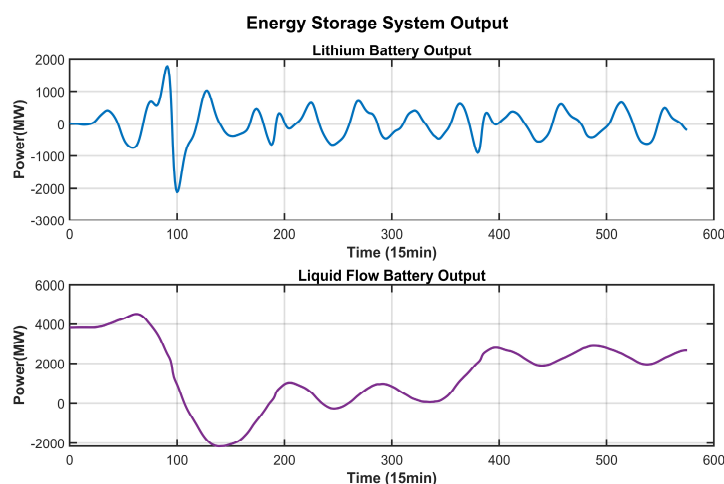


Figure 7. The Energy storage system output power.

Summing the power magnitudes of the high- and low-frequency components obtained from initial frequency division yields the total output of the energy storage system after frequency division. The maximum reconstruction error is 1.14×10^{-13} , and the root mean square error (RMSE) is 4.74×10^{-15} indicating that the frequency division results perfectly track the target storage output and exhibit exceptionally high consistency with the desired target output. This proves that the improved VMD frequency division method effectively achieves energy storage power allocation.

In terms of solving the site selection and capacity determination model, considering that this model is generally a large-scale mixed integer programming model (MIP), it is quite difficult to solve [22]. This paper conducts simulation verification based on Scheme 5 in the Case Setting to test the IWAA algorithm through multi-dimensional comparisons with advanced algorithms including COOT [23], PSO [25], DE [26], and WAA [27]. The comparative solving results are shown in Table 3. Results from Table 3 demonstrate that over 30 experimental trials: For penalty costs: the optimal solution obtained by IWAA is 13.3% lower than suboptimal WAA and 19.1% lower than worst-performing COOT; the mean value is 14.7% lower than DE; the standard deviation is 17.7% lower than PSO. For total costs: the optimal solution obtained by IWAA is 10.5% lower than suboptimal DE; the mean value is 7.2% lower than WAA; the standard deviation approaches that of optimal DE. These results indicate that IWAA demonstrates significant competitiveness in solution quality, stability, and convergence robustness, verifying its superiority in solving the proposed multi-objective bi-level planning problem.

Table 3. The solution results of 5 algorithms.

<i>z</i>		COOT	PSO	DE	WAA	IWAA
Penalty costs	Best	2.67×10^6	2.41×10^6	2.53×10^6	2.49×10^6	$2.16\text{E} \times 10^6$
	Avg.	3.12×10^7	$2.74\text{E}+07$	2.85×10^7	2.86×10^6	$2.43\text{E} \times 10^6$
	Std.	2.64×10^6	2.37×10^6	2.11×10^7	2.34×10^6	$1.95\text{E} \times 10^6$
Total costs	Best	9.64×10^6	9.97×10^6	9.59×10^6	9.06×10^6	$8.58\text{E} \times 10^6$
	Avg.	1.22×10^7	1.35×10^7	9.85×10^6	9.42×10^6	$8.74\text{E} \times 10^6$
	Std.	3.09×10^6	2.63×10^6	2.08×10^6	2.57×10^6	2.48×10^6

* Bolded entries in the table indicate optimal results.

Figures 8–12 display the scheduling schemes obtained by each algorithm. Each stacked histogram represents the output and input of all resources within a day, where positive and negative values indicate resource output and input, respectively, with

different colors denoting distinct resources. It demonstrates that the scheme derived from IWAA exhibits the lowest interaction frequency and minimal power exchange with the main grid. This confirms the algorithm's effectiveness in planning hybrid energy storage for regions with high renewable penetration, enhancing local renewable consumption capacity and strengthening grid stability.

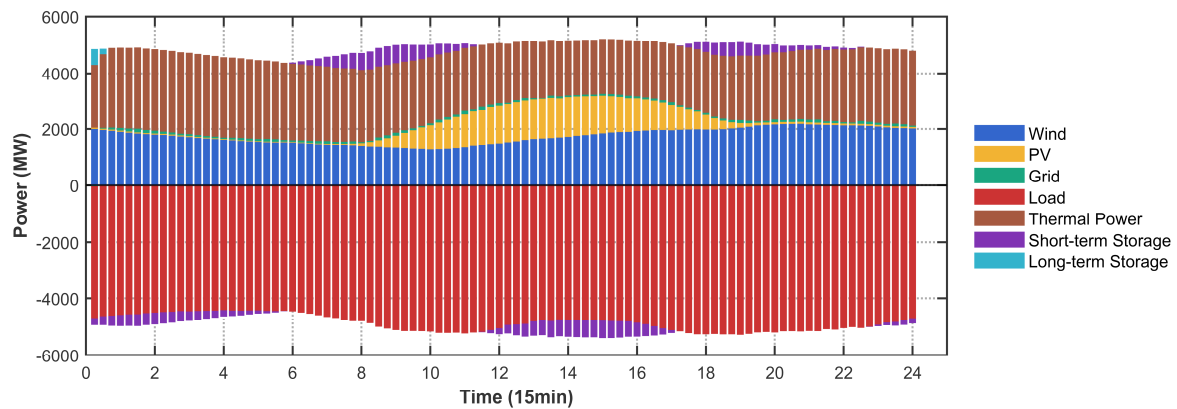


Figure 8. The planning result of COOT.

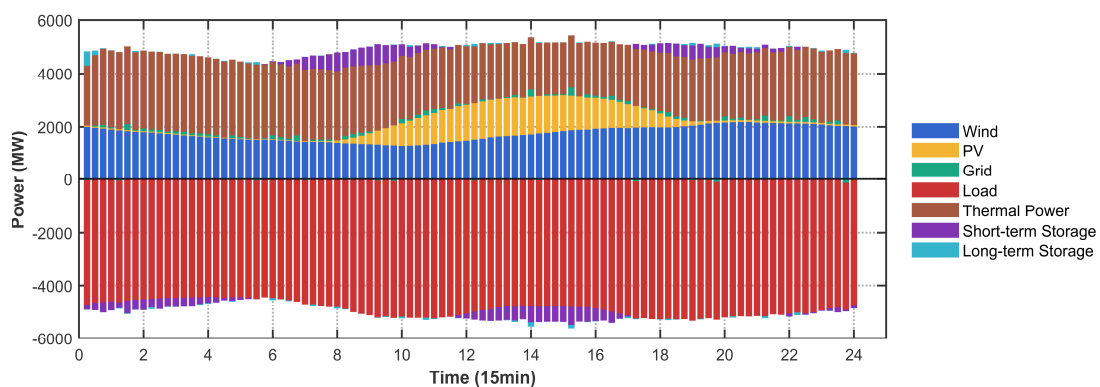


Figure 9. The planning result of PSO.

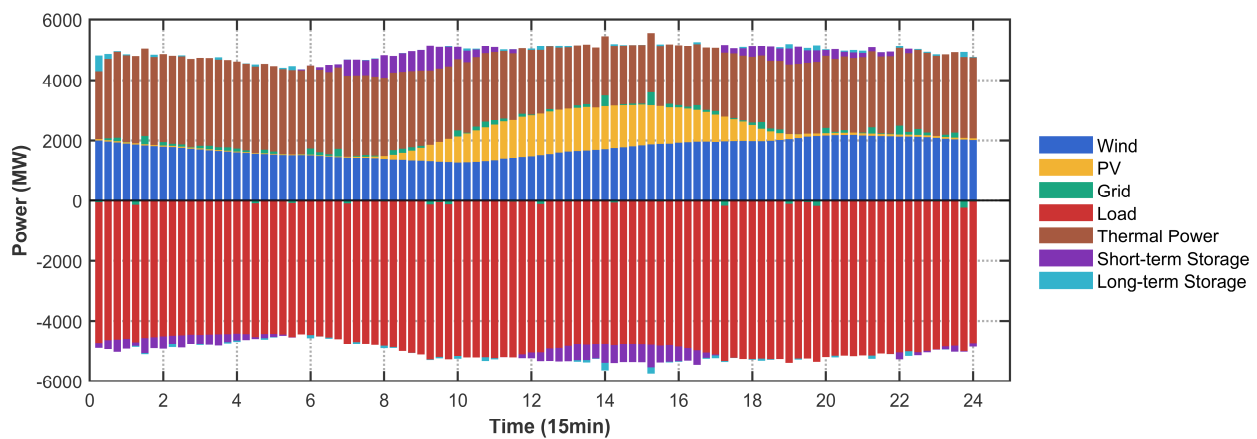


Figure 10. The planning result of DE.

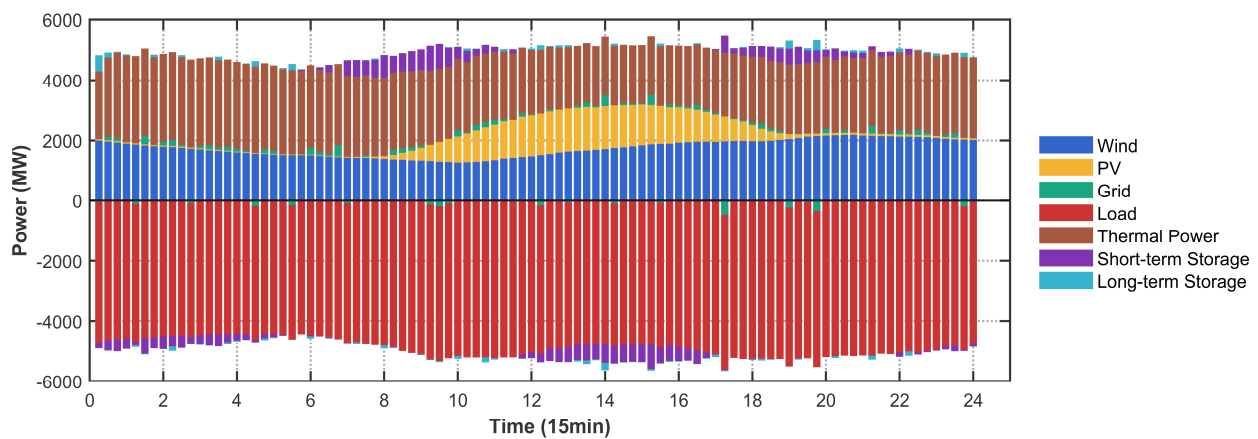


Figure 11. The planning result of WAA.

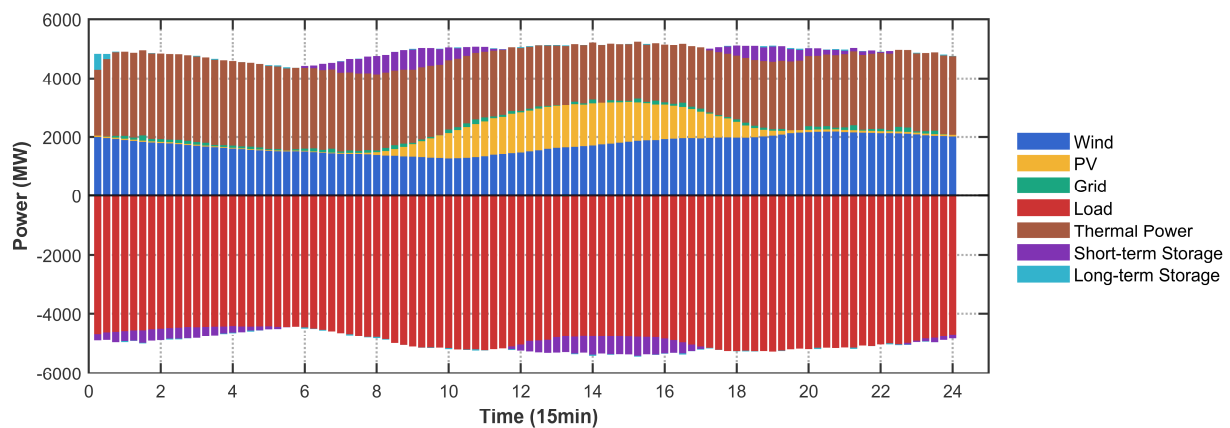


Figure 12. The planning result of IWAA.

Sensitivity Analysis

For grid operators, system stability is paramount, and the tolerance for flexibility shortfalls plays a key role in shaping energy storage planning decisions. In this section, we use the penalty coefficient λ_0 from the main text as the baseline, and set $\lambda \in \{0.1\lambda_0, 0.2\lambda_0, 0.3\lambda_0, 0.4\lambda_0, 0.5\lambda_0\}$ to reflect values commonly acceptable in operational practice. We then compare how different penalty ranges affect the planned storage capacity and total system cost. The results are presented in Table 4.

Table 4. Energy Storage Capacity Under Different Penalty Factors.

	Lithium-Ion Battery Capacity (MWh)	Lithium-Ion Battery Power (MW)	Flow Battery Capacity (MWh)	Flow Battery Power (MW)
$0.1\lambda_0$	3068	2484	6254	5175
$0.2\lambda_0$	2860	2333	5830	4860
$0.3\lambda_0$	2600	2160	5300	4500
$0.4\lambda_0$	2236	1901	4558	3960
$0.5\lambda_0$	1950	1685	3675	3110

As shown in the table, the capacities of both types of energy storage decrease as λ increases. When the penalty coefficient is raised from $0.1\lambda_0$ to $0.5\lambda_0$, lithium-ion battery capacity drops from 3068 MWh to 1950 MWh and rated power declines from 2484 MW to

1685 MW; flow battery capacity decreases from 6254 MWh to 3675 MWh and rated power from 5175 MW to 3110 MW. This indicates that a lower system tolerance leads to reduced storage capacity and power requirements, helping to decrease overall investment in storage. If grid operators prioritize system stability under extreme conditions, a penalty coefficient in the range of $0.2 \lambda_0$ to $0.3 \lambda_0$ can be selected to maintain higher redundancy in energy and power, delivering longer equivalent discharge durations. Conversely, in the presence of tighter budget constraints and acceptable flexibility shortfalls, a higher coefficient can be chosen to minimize investment, but this also increases operational risk. Across all scenarios, the capacity and power of flow batteries are significantly higher than those of lithium-ion batteries, highlighting the need for large-scale flow batteries to balance substantial low-frequency energy fluctuations within the system.

5. Conclusions

This study addresses the challenges of flexibility supply-demand balance under high-penetration renewable integration in power grids. It proposes a bi-level optimization planning model considering collaborative configuration of Hybrid Energy Storage Systems (HESS) and develops an IWAA Algorithm to efficiently approximate the pareto frontier for multi-objective problems. To validate the effectiveness and superiority of the proposed model and method, five distinct energy storage configuration schemes are designed in a typical regional grid scenario, with IWAA performance compared against other advanced optimization algorithms. Simulation results conclusively demonstrate the following advantages: (I) The proposed HESS bi-level planning scheme optimizes storage siting/sizing and power dynamic allocation, achieving integrated minimization of storage investment and operation-maintenance costs, flexibility penalty costs, and system losses. (II) The proposed improved VMD algorithm incorporating PSO successfully achieves precise decomposition of total target storage output, providing reliable technical support for complementary HESS operation and mitigating multi-timescale fluctuations. (III) The proposed IWAA significantly enhances global exploration capability in solving complex nonlinear planning problems. Multi-round comparative experiments confirm IWAA's superiority in optimal solution quality, mean solution quality, and stability when solving this bi-level planning model.

The proposed HESS collaborative planning framework and IWAA solver effectively optimize storage configuration and operation strategies in high-renewable grids, significantly reducing total system costs, enhancing renewable consumption capacity, and strengthening grid reliability and operational stability. It is important to note that this study focuses on capacity and power constraints while neglecting battery energy attenuation degradation during operation. While this simplification is common in planning-level studies to manage computational complexity, future work will incorporate detailed, operation-dependent degradation models for both Li-ion and flow batteries. This enhancement will provide a more accurate lifecycle cost analysis and refine the understanding of the long-term economic viability of HESS in renewable-rich grids. At last, the present model does not implement advanced risk constraints nor multi-stage drivers of long-term structural uncertainty. These aspects are important and will be addressed in future work with a multi-stage stochastic extension, building on the present scenario construction.

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